Change detectives: what became something new?

Teacher guide and worked answers

Describe burning and rusting as changes that form new materials, and distinguish them from a change of state.

40 minutes. Use Lesson.html for interactive teaching or Lesson.pptx for editable slides. Slides.pdf is the static alternative. Select one pupil response route and print the shared evidence it requires. Detailed answers below are for staff.

Scheme of work

_passsg/inputs/GROW SOW 2026-27.xlsx | GROW Weekly - Summer | C35

Describe irreversible changes (burning, rusting, new materials formed).

Directly develops the stated outcome. Original teacher-authored exemplar; route support is responsive and not a qualification or fixed ability label.

Prior learning

Describe melting, freezing and dissolving.

Explain how salt can be recovered from a solution.

Success evidence

Use evidence to distinguish a new substance from a change of shape or state.

Explain why cooling burnt wood or rust does not restore the starting material.

Describe iron rusting and fuel burning without claiming that matter disappears.

Equipment

Shared evidence sheet, one selected response route and exit sheet; pencils.

Preparation

Read the worked answers and preview the three interactive decisions.

Print shared page 1, one route from pages 2-4, and page 5.

Practical care

This lesson uses supplied evidence cards and diagrams. No flames, hot wax, chemicals or rusty sharp objects are needed.

If using a sample for looking, the teacher selects a clean sealed display; pupils do not scrape or handle corroded edges.

Ready to teach alternative

All observations are supplied as constructed teaching cases. The complete lesson is a desk investigation; pupils are not asked to make or burn materials.

Teaching sequence

Slide	Stage	Minutes	Focus
1	Opening	0-2	The workshop has mixed up its labels
2	Retrieval	2-5	Three things we can recover
3	I do	5-13	I ask two different questions
4	I do	Within stage	I explain the rusty bracket
5	I do	Within stage	I separate two candle processes
6	We do	13-21	The evidence sorting board
7	We do	Within stage	Repair the workshop rule
8	We do	Within stage	Two labels for one candle
9	Hinge	21-24	Which is the strongest evidence?
10	You do	24-34	Write a workshop diagnosis
11	You do	Within stage	Use a cause, not only a description
12	You do	Within stage	Check the two-change trap
13	Review	34-38	Audit the diagnosis
14	Review	Within stage	Choose the next exhibit
15	Exit	38-40	Leave a reliable label

Times are a suggested 40-minute route. Slides marked Within stage share the preceding stage allocation. The optional HTML timer starts only when selected.

1 Opening The workshop has mixed up its labels

Frame a desk evidence investigation. Do not classify by dramatic appearance. Point to the difference between an object and the substance it is made from.

2 Retrieval Three things we can recover

R1 cool below freezing point. R2 no; it remains in solution and can be recovered. R3 no; state changes retain substance identity. Use responses to recap the preceding recovery lesson briefly.

3 I do I ask two different questions

Think aloud: I can freeze melted ice; the substance is water throughout. A torn paper sheet is hard to restore, yet tearing has not by itself created a new substance. This is why “hard to undo” alone is an unreliable chemical test. Do not teach all physical changes as reversible.

4 I do I explain the rusty bracket

Shared I do time. Use the word model as a relationship, not a particle picture or balanced equation. Rust is a mixture of iron oxides/hydroxides, often simplified as hydrated iron oxide; that name is not required. Ordinary rusting needs both oxygen and water. Salt can accelerate it but is not essential.

5 I do I separate two candle processes

Shared I do time. Solid wax melts; liquid wax is drawn up a wick and vaporises before reacting. The top and bottom panels represent different changes, not a single arrow joining liquid wax directly to products. Complete combustion gives carbon dioxide and water; incomplete combustion can also produce soot and other products. No live demonstration.

6 We do The evidence sorting board

Use 3 minutes of We do: A/B physical; C/D chemical. For B explain same paper even though sheet restoration is difficult. C has analysis evidence; D identifies products. Each pupil commits a sort by pointing or noting letters before feedback.

7 We do Repair the workshop rule

Use 2 minutes: torn paper is a counterexample. Better question: has evidence shown a new substance formed? C rust and D burning support it. A visual change is a clue, not sufficient proof of a chemical change.

8 We do Two labels for one candle

Use 3 minutes: E-A physical melting; E-B chemical burning. Cooling can solidify remaining liquid wax. Burnt wax reacts with oxygen, making products including gases; atoms do not vanish.

Reveal: Melting: state change. Burning: new substances, including gases.

9 Hinge Which is the strongest evidence?

A incorrect: shape can change physically. B correct: new-substance evidence. C incorrect: physical changes can be difficult to undo. If uncertain, return to torn paper and rusty iron, asking about substance identity.

A. It has a different shape. - A shape change can leave the substance unchanged.

B. Tests identify a new substance. - Formation of a new substance defines a chemical change.

C. It is awkward to put back. - Torn paper can be awkward to restore without a chemical change.

10 You do Write a workshop diagnosis

Independent 10 minutes over three slides. Select support from hinge evidence. Offer the evidence sheet on all routes. Adult may read questions but must not provide classification or causal explanation.

11 You do Use a cause, not only a description

Shared You do time. Prompt the type of explanation without naming answers. Pupils can use labels and arrows instead of a paragraph.

12 You do Check the two-change trap

Shared You do time. Check task access only; do not read model answers until the independent attempt is complete.

13 Review Audit the diagnosis

Review after collecting independent evidence. Accept scientific meaning rather than prescribed wording. A correctly classified case without a reason is partial, not secure evidence.

14 Review Choose the next exhibit

Lundy pupil voice: collect suggestions, select one improvement, and explain how pupil reasoning affected the teacher’s next explanation. Passing or submitting a written suggestion is valid.

15 Exit Leave a reliable label

E1 new rust material forms from iron with oxygen/water. E2 no new substance just from tearing. E3 reaction forms new products, unlike state change. Collect before revealing answers.

Answers and assessment

R1-R3

R1 cool water below its freezing point. R2 salt remains present in solution and can be recovered. R3 no, melting is a state change.

W1-W3

W1 A/B: no new substance, physical; C/D: new substances, chemical. W2 torn paper retains the same material even though the sheet is hard to restore; ask whether new substances formed. W3 E-A physical melting, E-B chemical burning; cooling can solidify remaining melted wax, but does not turn combustion products back into wax. Gaseous products enter the air.

Hinge A-C

A incorrect: a shape change may be physical. B correct: new substance identified. C incorrect: difficulty reversing does not establish chemical change.

S1-S5

S1 new material. S2 oxygen/water/rust. S3 substance still water. S4 tearing changes shape/size but does not itself make a new substance. S5 melting: solid to liquid, same wax, cooling reverses it; burning: wax vapour reacts with oxygen to make new substances such as carbon dioxide and water.

N1-N4

N1 ice physical/same water; paper physical/same paper; iron chemical/new rust; wood chemical/new ash and gases. N2 rust is a chemically different material; cooling does not undo that reaction. N3 melting changes wax’s state; burning wax vapour reacts with oxygen and produces new substances. N4 gases spread into the surrounding air; the open tray does not collect all products.

T1-T3

T1 torn paper is a valid counterexample; a physical change can be hard to reverse. Look for formation of new substances. T2 gases escaped and oxygen entered from the air as a reactant. Matter was rearranged, not destroyed. The original wood mass excludes the incoming oxygen, so total gas product mass cannot be obtained as $10-2$ alone. T3 separate accurate labels: melting is state change and liquid wax can solidify; burning makes new products and cooling does not restore wax. Accept accurate mention of incomplete-combustion products.

E1-E4 / reflection

E1 rusting forms new substances as iron reacts with oxygen and water. E2 same paper material remains. E3 melting changes state, burning forms new substances. E4 into the surrounding air as gases; ash may remain. Reflection accepts any correctly justified correction.

Evidence and responsive teaching

Secure core evidence: classifies both rusting and burning using new-substance reasoning and distinguishes melting from burning. Developing: classification is correct but relies only on appearance or difficulty reversing. Re-teach with the ice/paper contrast and the two separate candle processes. Stretch open-system mass reasoning is optional and must not become a core barrier.

Teaching and assessment notes

40 minutes: opening 2; retrieval 3; I do 8; We do 8; hinge 3; You do 10; review 4; exit 2. Zero-minute slides share their stage time.

Print shared page 1, ONE selected response route (2, 3 or 4), and exit page 5. Do not require all routes. Adult scribing records the pupil's own words.

The SOW term "irreversible" refers to everyday classroom reversal, not a universal rule that all chemical reactions are irreversible. Do not extend the shortcut to torn paper.

Salt is not essential for rusting; ordinary rusting requires oxygen and water. The lesson uses supplied analysis rather than treating a colour change alone as proof.

All masses and workshop events are constructed illustrations, not collected school observations. No home burning task is suggested.

Access and responsive teaching

Supported

Use short choices, named evidence cards and a sentence bank. Accept pointing followed by one spoken reason.

Standard

Classify four cases and write an evidence-based repair label explaining new substances.

Stretch

Evaluate a misleading irreversible-change rule and explain an open-system mass observation qualitatively.

Regulation

Use fictional workshop cases. Offer private responses, drawing or scribing; pupils may pass and return to a shared question. No sensory smoke or flame exposure is part of the task.

Misconceptions to check

Anything difficult to undo is a chemical change.

Tearing paper changes its form without necessarily making a new substance. New-substance evidence is the key.

Check: Ask whether cutting paper creates a new substance.

Wax melting and wax burning are the same change.

Melting changes state; wax vapour burning reacts with oxygen to make new substances.

Check: Ask what cooling can reverse.

Rust is iron with brown dirt on it.

Rust contains new substances formed as iron reacts with oxygen and water.

Check: Ask whether wiping or cooling turns rust into iron.

Burnt material disappears.

Matter is rearranged into products, including gases that can escape into air.

Check: Ask what an open tray fails to collect.

Word	Meaning
substance	A material with a particular chemical identity.
physical change	A change of form or state that does not make a new substance.
chemical change	A change in which new substances are formed.
rusting	Iron reacting with oxygen and water to form rust.
burning	A chemical reaction of a fuel with oxygen that transfers energy.
irreversible	Not returned to the starting material by simply reversing the classroom process.

Scientific references

ACS: What is a chemical reaction?

Checked September 2026. New substances, combustion products and conservation of atoms; our card investigation and diagrams are original. No source practical is required.

OpenStax Chemistry 2e: Physical and chemical properties

Checked September 2026. Physical versus chemical change, including corrosion; terminology is adapted to the GROW scheme.

DfE: Science programmes of study

Checked September 2026. Upper-KS2 changes of materials and new-material outcomes; no accreditation claim.